Table 1. Measured soil properties of the DI field

Depth (cm)	Sand (%)	Silt (%)	Clay (%)	Bulk density (g cm⁻³)	Texture
0–10	29.70	59.66	10.64	1.65	Silt loam
10–20	30.81	58.67	10.52	1.65	Silt loam
20–40	19.43	68.25	12.32	1.69	Silt loam
40–60	13.22	70.58	16.20	1.29	Silt loam
60–80	10.38	74.87	14.75	1.48	Silt loam
80–100	11.81	76.53	11.66	1.45	Silt loam

Note. Source: Wang (2023), Table 2-1. Particle fractions: >0.05 mm, 0.002–0.05 mm and <0.002 mm.

Table 2. Annual DI data and calibration targets

Year	Irrigation (mm)	Observed yield (t ha⁻¹)	CC / B pairs	ET reference (mm)	Es target (mm)
2019	445.27	16.55	12 / 9	661.45	76
2020	559.76	17.29	13 / 10	637.03	79
2021	633.47	15.86	11 / 8	626.81	78
2022	404.84	14.08	12 / 10	584.11	81

Note. Yield is at 13% moisture. Es is the archived approximate/reconstructed calibration target. The original manuscript label “Es,obs” is clarified here; numerical values are unchanged. Event-sum irrigation for 2021 is 633.47 mm.

Table 3. Calibrated parameters and common independent Sobol ranges

Parameter	Phase	Meaning	Unit	Sobol range
CCx	1	Maximum canopy cover	fraction	0.6–0.99
CGC_CD	1	Canopy growth coefficient	d-1	0.08–0.16
CDC_CD	1	Canopy decline coefficient	d-1	0.03–0.12
SenescenceCD	1	Senescence onset	d	115–145
Kcb	1	Basal crop coefficient	dimensionless	1.05–1.6
WP_early	2	Early effective water productivity	g m-2	10–40
WP_late	2	Late effective water productivity	g m-2	10–40
HI0	2	Reference harvest index	fraction	0.35–0.6

Note. These are common independent uniform Sobol ranges. Annual MCMC prior families and supports are separately archived.

Table 4. Phase-specific retained means used in deterministic simulations

Parameter	Unit	2019	2020	2021	2022
CCx	fraction	0.9771	0.9773	0.9594	0.9374
CGC_CD	d^{-1}	0.0988	0.0956	0.1154	0.1318
CDC_CD	d^{-1}	0.0603	0.0607	0.0663	0.0642
SenescenceCD	d	137.38	137.84	134.85	134.95
Kcb	–	1.4866	1.4352	1.4161	1.3775
WP_early	g m^{-2}	18.204	11.255	10.877	13.860
WP_late	g m^{-2}	30.585	25.849	21.938	31.498
HI0	fraction	0.4124	0.5738	0.5800	0.4013

Note. Recalculated from 20 complete original checkpoints using the archived phase-specific burn-in. Phase 2 is conditional on phase 1.

Table 5. Calibration performance for canopy cover, biomass and grain yield

Year	CC R^2cal	CC RMSE	Biomass R^2cal	B RMSE (t ha^{-1})	Observed Y (t ha^{-1})	Simulated Y (t ha^{-1})	Yield RE (%)
2019	0.9932	0.0316	0.9926	1.0498	16.55	16.702	+0.92
2020	0.9794	0.0482	0.9766	1.1393	17.29	16.804	-2.81
2021	0.9902	0.0370	0.9685	1.2520	15.86	15.197	-4.18
2022	0.9306	0.0596	0.9966	0.5562	14.08	14.078	-0.02

Note. R^2cal = 1 − SSE/SST. CC is a fraction; biomass is aboveground dry matter. These are calibration comparisons, not new independent validation observations.

Table 6. Four-year descriptive yield sensitivity under common input ranges

Parameter	Mean S_1	Mean S_t	Minimum S_t	Maximum S_t
WP_late	0.377	0.396	0.353	0.426
HI0	0.193	0.206	0.203	0.208
WP_early	0.142	0.164	0.139	0.198
Kcb	0.130	0.145	0.139	0.148
CCx	0.049	0.057	0.052	0.060
CGC_CD	0.033	0.040	0.029	0.051
SenescenceCD	0.003	0.011	0.006	0.015
CDC_CD	0.003	0.006	0.003	0.009

Note. Each annual value averages three N=512 scrambles; the four annual values receive equal weight. Total-effect indices need not sum to one.

Table 7. Selected five-stage NSGA-III schedules at 95% observed yield

Year	Recorded I (mm)	Selected I (mm)	Selected yield (t ha^{-1})	I reduction (%)	Yield RE (%)	ET (mm)
2019	445.27	209.83	15.725	52.88	-4.99	599.84
2020	559.76	306.50	16.427	45.24	-4.99	592.92
2021	633.47	264.50	15.068	58.25	-4.99	593.68
2022	404.84	257.47	13.390	36.40	-4.90	568.93

Note. The candidate minimizes actual irrigation within the feasible nondominated union of five formal NSGA-III final populations. Yield RE uses published observed yield.

Table 8. Water-balance decomposition of irrigation reduction

Year	I reduction (mm)	ET decrease (mm)	Lower final S (mm)	Other losses (mm)	Storage share (%)
2019	235.44	38.16	193.14	4.14	82.03
2020	253.26	17.04	234.87	1.34	92.74
2021	368.97	11.46	356.85	0.66	96.72
2022	147.37	27.85	119.50	0.02	81.09

Note. Differences are recorded DI minus selected candidate. Storage integrates the unchanged 0–2.4 m simulated profile. Signed components sum to irrigation reduction.

Table 9. Five-stage actual irrigation totals and event counts

Year	Seedling (mm)	Jointing (mm)	Heading (mm)	Grain filling (mm)	Maturity (mm)	Positive events	Events ≥0.1 mm
2019	85.168	4.645	119.752	0.241	0.025	12	8
2020	108.288	94.704	2.255	88.711	12.541	13	8
2021	0.003	178.782	0.054	85.634	0.027	9	3
2022	0.035	181.420	2.935	73.030	0.044	8	6

Note. All exact positive depths count as events after same-day merging; ≥0.1 mm counts are additional statistics.
Unrounded controls and numerical traces are retained.

Table 10. Final feasible-population union metrics from five seeds

Year	NSGA-III min f (mm)	NSGA-II min f (mm)	III minus II (mm)	NSGA-III HV	NSGA-II HV
2019	209.832	209.866	-0.035	0.541654	0.541554
2020	306.499	305.027	+1.472	0.456772	0.459376
2021	264.500	264.546	-0.046	0.571337	0.571307
2022	257.465	256.765	+0.700	0.377835	0.379541

Note. Both algorithms share disclosed initialization anchors, objectives and constraints. HV uses the original annual normalization and reference point (0, 1.01).